

FCC PART 15 SUBPART B TEST REPORT

For Electromagnetic Interference of

Report Reference No.: ATSE240501511

Date of issue.....: 2024-06-04

Testing Laboratory.....: ATS Electronic Technology Co., Ltd.

Address: 3/F. of Building A, 1/F. of Building B & 1/F. of Building C, No.1,
Hedong 3rd Road, Jinxia Community, Changan, Donguan,
Guangdong, China

Applicant's name: Shandong Goldencell Power Technology Co.,Ltd.

Address: No.X6699 Guangming Boulevard, Hightech District Zaozhuang
City Shandong Province

Test specification:

Test item description.....: UPS uninterruptible power supply

Model/Type reference.....: UPS 1000VA Pro/800W

Ratings: I/P: 120Vac, 50/60Hz, 1000VA/800W

O/P: 120Vac, 60Hz

Battery specification: 25.6Vdc, 9Ah, 230Wh.

Responsible Engineer



(Brant Yang / Engineer)

Approved by



(Simon Zeng / EMC Manager)

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1. CERTIFICATION

Testing Laboratory: ATS Electronic Technology Co., Ltd.

Address.....: 3/F. of Building A, 1/F. of Building B & 1/F. of Building C, No.1,
Hedong 3rd Road, Jinxia Community, Changan, Donguan,
Guangdong, China

Applicant's name: Shandong Goldencell Power Technology Co.,Ltd.

Address.....: No.X6699 Guangming Boulevard, Hightech District Zaozhuang
City Shandong Province

Manufacturer: Same as applicant

Address.....: Same as applicant

Factory.....: Same as applicant

Address.....: Same as applicant

Test specification:

Test item description.....: UPS uninterruptible power supply

Trade Mark: The logo for GOLDELMATE features the word 'GOLDELMATE' in a bold, orange, sans-serif font, with a stylized lightning bolt graphic integrated into the letter 'M'.

Model/Type reference.....: UPS 1000VA Pro/800W

Test Sample.....: UPS 1000VA Pro/800W

Tested Power.....: 120Vac, 60Hz

Standards: FCC Part 15 Subpart B

The device described above was tested by ATS Electronic Technology Co., Ltd. to determine the maximum emission levels emanated from the device and severity levels of the device endure and it performance criterion. The measurement results are contained in this test report and ATS Electronic Technology Co., Ltd. assumes full responsibility for the accuracy and completeness of these measurements. This report shows the EUT is technically compliance with the above official standards. This report applies to the above sample only and shall not be reproduced in part without written approval of ATS Electronic Technology Co., Ltd.

2. SUMMARY OF TEST RESULTS

Test procedures according to the technical standards:

EMC Emission				
Standard	Test Item	Limit	Judgment	Remark
FCC Part 15 B	Conducted Emission	Clause 15.107	PASS	
	Radiated Emission	Clause 15.109	PASS	

2.1 MEASUREMENT UNCERTAINTY

The reported uncertainty of measurement $y \pm U$, where expanded uncertainty U is based on a standard uncertainty multiplied by a coverage factor of $k=2$, providing a level of confidence of approximately 95 %.

A. Conducted Measurement :

Test Site	Method	Measurement Frequency Range	U , (dB)	NOTE
C01	ANSI	150 KHz ~ 30MHz	2.44	

B. Radiated Measurement :

Test Site	Method	Measurement Frequency Range	Ant. H / V	U , (dB)	NOTE
R03	ANSI	30MHz ~ 200MHz	V	3.42	
		30MHz ~ 200MHz	H	3.52	
		200MHz ~ 1,000MHz	V	3.52	
		200MHz ~ 1,000MHz	H	3.54	

2.2 DESCRIPTION OF TEST MODES

To investigate the maximum EMI emission characteristics generates from EUT, the test system was pre-scanning tested base on the consideration of following EUT operation mode or test configuration mode which possible have effect on EMI emission level. Each of these EUT operation mode(s) or test configuration mode(s) mentioned above was evaluated respectively.

Pretest Mode	Description
Mode 1	Charging+Working
Mode 2	Working
Mode 3	Charging

For Conducted Test	
Final Test Mode	Description
Mode 1	Charging+Working
Mode 2	Charging

For Radiated Test	
Final Test Mode	Description
Mode 1	Charging+Working
Mode 2	Working
Mode 3	Charging

3. EMC EMISSION TEST

3.1 CONDUCTED EMISSION MEASUREMENT

3.1.1 POWER LINE CONDUCTED EMISSION (Frequency Range 150KHz-30MHz)

FREQUENCY (MHz)	Class B (dBuV)	
	Quasi-peak	Average
0.15 -0.5	66 - 56 *	56 - 46 *
0.50 -5.0	56.00	46.00
5.0 -30.0	60.00	50.00

Note:

- (1) The tighter limit applies at the band edges.
- (2) The limit of " * " marked band means the limitation decreases linearly with the logarithm of the frequency in the range.

3.1.2 MEASUREMENT INSTRUMENTS LIST

Item	Kind of Equipment	Manufacturer	Type No.	Serial No.	Calibrated until
1	EMI Test Receiver	R&S	ESCI	101569	10/20/2024
2	LISN	Schaffner	MN2050D	1467	10/20/2024
3	LISN	R&S	ENV216	101348	10/20/2024

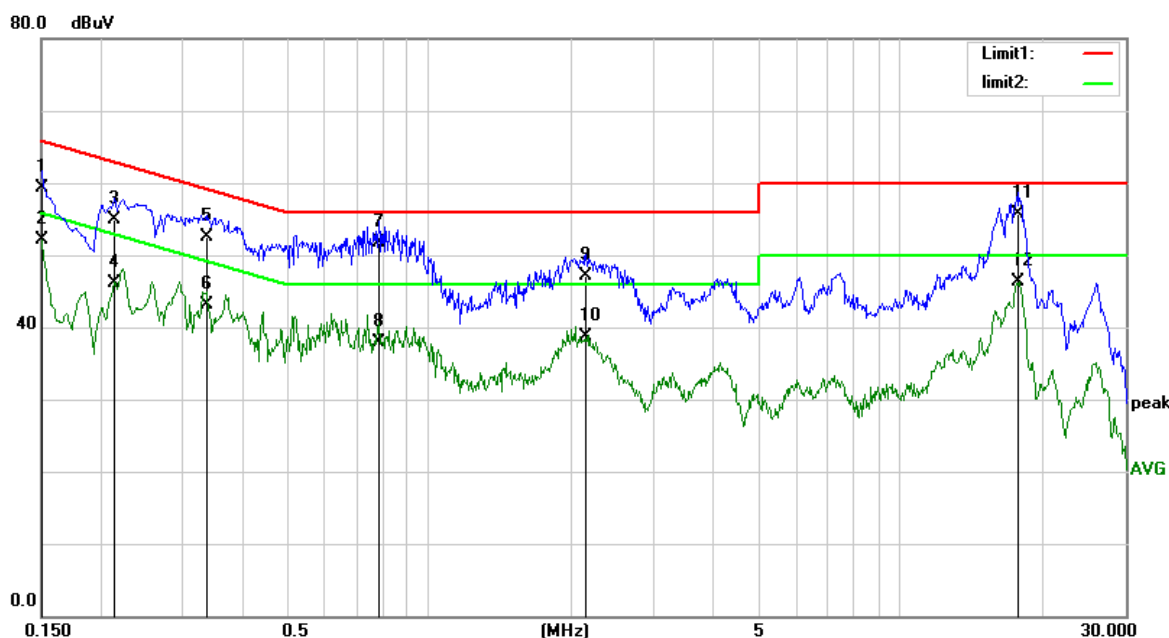
3.1.7 TEST RESULTS

EUT :	UPS uninterruptible power supply
Model No. :	UPS 1000VA Pro/800W
Test Mode :	Charging, Charging+Working
Test Result:	PASS

Remark

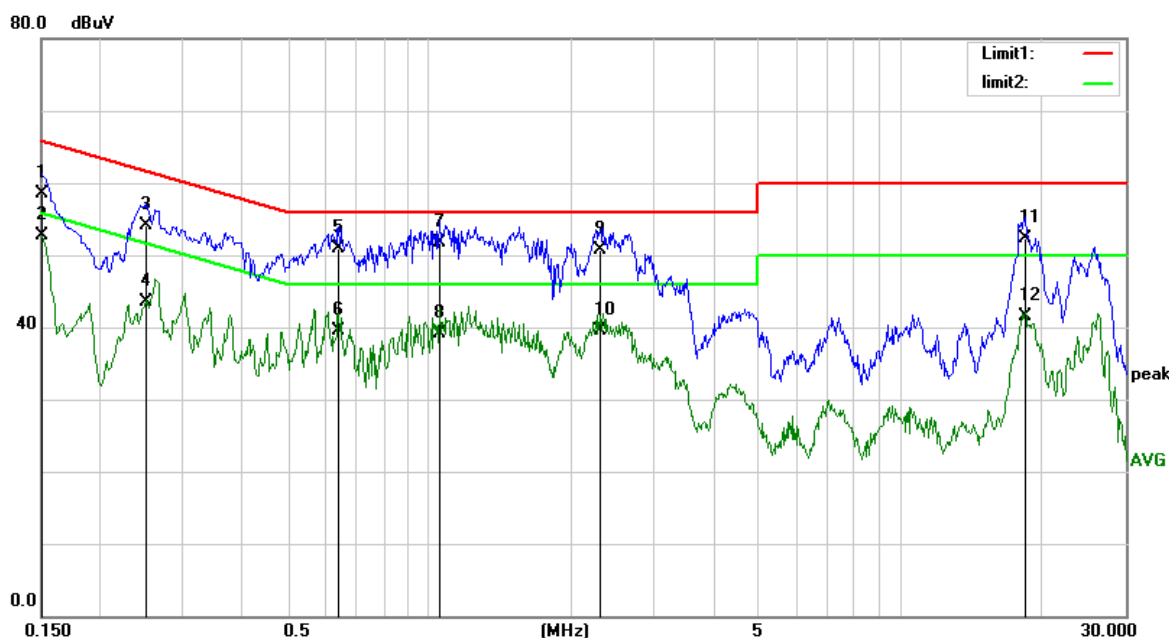
- (1) Reading in which marked as QP means measurements by using are Quasi-Peak Mode with Detector BW=9KHz ; SPA setting in RBW=10KHz,VBW =10KHz, Sweep. Time = 0.3 sec./MHz. Reading in which marked as AV means measurements by using are Average Mode with instrument setting in RBW=1MHz,VBW=10Hz, Sweep. Time =0.3 sec./MHz.
- (2) All readings are QP Mode value unless otherwise stated AVG in column of 『Note』 . If the QP Mode Measured value compliance with the QP Limits and lower than AVG Limits, the EUT shall be deemed to meet both QP & AVG Limits and then only QP Mode was measured, but AVG Mode didn't perform. In this case, a “ * ” marked in AVG Mode column of Interference Voltage Measured.
- (3) Measuring frequency range from 150KHz to 30MHz.
- (4) Measurement result=Reading + Correct.

EUT:	UPS uninterruptible power supply	Model No. :	UPS 1000VA Pro/800W
Temperature:	24 °C	Relative Humidity:	55 %
Phase:	L1	Test Power :	120Vac, 60Hz
Standard:	(CE)FCC PART 15 class B_QP	Test By:	Devil
Test Mode :	Charging		



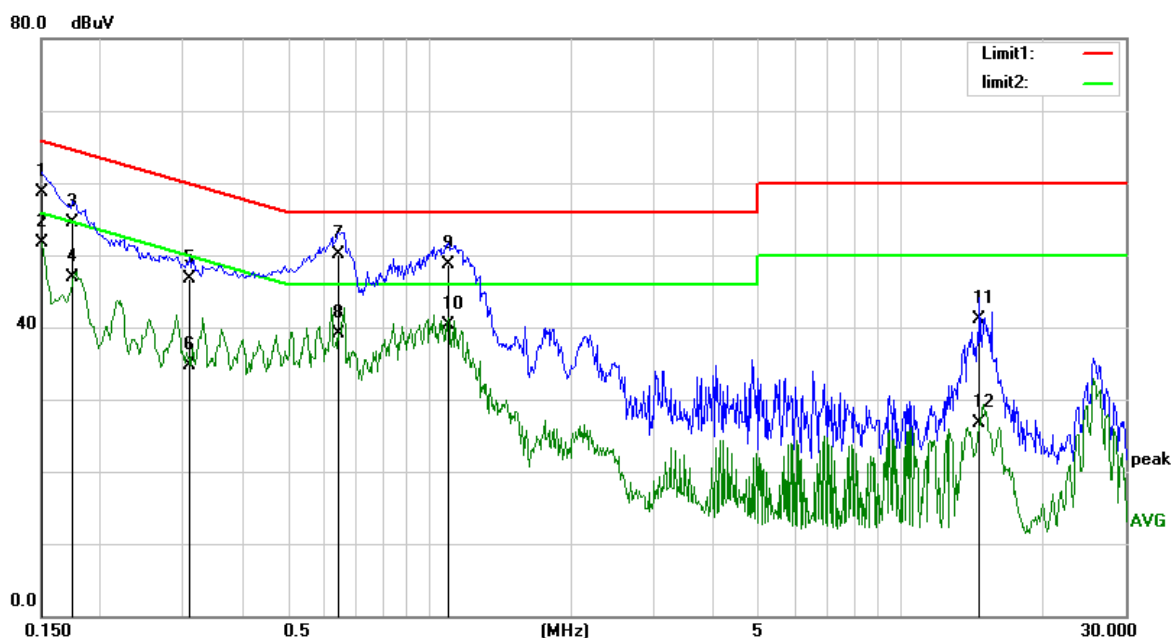
No. Mk.	Freq. MHz	Reading Level dBuV	Correct Factor dB	Measure- ment dBuV	Limit dB	Over dB	Detector	Comment
1	0.1500	48.96	10.30	59.26	66.00	-6.74	QP	
2	0.1500	41.75	10.30	52.05	56.00	-3.95	AVG	
3	0.2139	44.58	10.31	54.89	64.17	-9.28	QP	
4	0.2139	35.84	10.31	46.15	54.17	-8.02	AVG	
5	0.3379	42.28	10.20	52.48	60.63	-8.15	QP	
6	0.3379	32.97	10.20	43.17	50.63	-7.46	AVG	
7	0.7820	41.52	10.18	51.70	56.00	-4.30	QP	
8	0.7820	27.73	10.18	37.91	46.00	-8.09	AVG	
9	2.1499	36.76	10.26	47.02	56.00	-8.98	QP	
10	2.1499	28.43	10.26	38.69	46.00	-7.31	AVG	
11	17.7099	45.34	10.34	55.68	60.00	-4.32	QP	
12 *	17.7099	35.90	10.34	46.24	50.00	-3.76	AVG	

EUT:	UPS uninterruptible power supply	Model No. :	UPS 1000VA Pro/800W
Temperature:	24 °C	Relative Humidity:	55 %
Phase:	N	Test Power :	120Vac, 60Hz
Standard:	(CE)FCC PART 15 class B_QP	Test By:	Devil
Test Mode :	Charging		



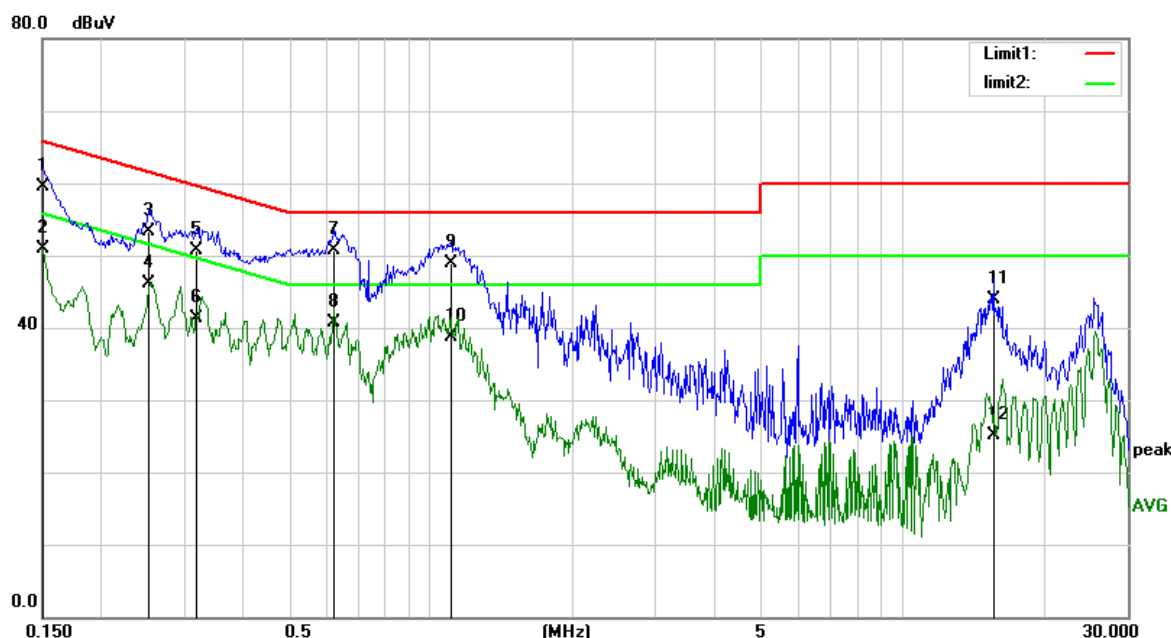
No. Mk.	Freq.	Reading Level	Correct Factor	Measure-ment	Limit	Over	Detector	Comment
	MHz	dBuV	dB	dBuV	dB	dB		
1	0.1500	48.29	10.22	58.51	66.00	-7.49	QP	
2 *	0.1500	42.44	10.22	52.66	56.00	-3.34	AVG	
3	0.2500	44.06	10.13	54.19	63.14	-8.95	QP	
4	0.2500	33.39	10.13	43.52	53.14	-9.62	AVG	
5	0.6419	40.50	10.32	50.82	56.00	-5.18	QP	
6	0.6419	29.12	10.32	39.44	46.00	-6.56	AVG	
7	1.0540	41.31	10.33	51.64	56.00	-4.36	QP	
8	1.0540	28.68	10.33	39.01	46.00	-6.99	AVG	
9	2.3060	40.58	10.21	50.79	56.00	-5.21	QP	
10	2.3060	29.31	10.21	39.52	46.00	-6.48	AVG	
11	18.3618	41.85	10.38	52.23	60.00	-7.77	QP	
12	18.3618	31.06	10.38	41.44	50.00	-8.56	AVG	

EUT:	UPS uninterruptible power supply	Model No. :	UPS 1000VA Pro/800W
Temperature:	24 °C	Relative Humidity:	55 %
Phase:	L1	Test Power :	120Vac, 60Hz
Standard:	(CE)FCC PART 15 class B_QP	Test By:	Devil
Test Mode :	Charging+Working		



No. Mk.	Freq.	Reading Level	Correct Factor	Measurement	Limit	Over	Detector	Comment
	MHz	dBuV	dB	dBuV	dB	dB		
1	0.1500	48.34	10.30	58.64	66.00	-7.36	QP	
2 *	0.1500	41.31	10.30	51.61	56.00	-4.39	AVG	
3	0.1737	44.17	10.31	54.48	65.32	-10.84	QP	
4	0.1737	36.56	10.31	46.87	55.32	-8.45	AVG	
5	0.3099	36.56	10.17	46.73	61.43	-14.70	QP	
6	0.3099	24.44	10.17	34.61	51.43	-16.82	AVG	
7	0.6419	39.82	10.34	50.16	56.00	-5.84	QP	
8	0.6419	28.84	10.34	39.18	46.00	-6.82	AVG	
9	1.0940	38.43	10.26	48.69	56.00	-7.31	QP	
10	1.0940	30.07	10.26	40.33	46.00	-5.67	AVG	
11	14.7260	30.83	10.30	41.13	60.00	-18.87	QP	
12	14.7260	16.32	10.30	26.62	50.00	-23.38	AVG	

EUT:	UPS uninterruptible power supply	Model No. :	UPS 1000VA Pro/800W
Temperature:	24 °C	Relative Humidity:	55 %
Phase:	N	Test Power :	120Vac, 60Hz
Standard:	(CE)FCC PART 15 class B_QP	Test By:	Devil
Test Mode :	Charging+Working		



No.	Mk.	Freq.	Reading Level	Correct Factor	Measurement	Limit	Over	Detector	Comment
		MHz	dBuV	dB	dBuV	dB	dB		
1		0.1500	49.20	10.22	59.42	66.00	-6.58	QP	
2	*	0.1500	40.69	10.22	50.91	56.00	-5.09	AVG	
3		0.2519	43.13	10.15	53.28	63.09	-9.81	QP	
4		0.2519	35.90	10.15	46.05	53.09	-7.04	AVG	
5		0.3183	40.37	10.31	50.68	61.19	-10.51	QP	
6		0.3183	30.93	10.31	41.24	51.19	-9.95	AVG	
7		0.6260	40.37	10.35	50.72	56.00	-5.28	QP	
8		0.6260	30.45	10.35	40.80	46.00	-5.20	AVG	
9		1.1060	38.55	10.27	48.82	56.00	-7.18	QP	
10		1.1060	28.51	10.27	38.78	46.00	-7.22	AVG	
11		15.5780	33.65	10.34	43.99	60.00	-16.01	QP	
12		15.5780	14.76	10.34	25.10	50.00	-24.90	AVG	

3.2 RADIATED EMISSION MEASUREMENT

3.2.1 LIMITS OF RADIATED EMISSION MEASUREMENT (Below 1000MHz)

FREQUENCY RANGE OF RADIATED MEASUREMENT (For FCC)

☒ FCC Class B Limit at 3m

Frequency	Distance	Field Strength	
MHz	Meter	$\mu\text{V/m}$	$\text{dB}\mu\text{V/m}$
30 to 88	3	100	40.0
88 to 216	3	150	43.5
216 to 960	3	200	46.0
Above 960	3	500	54.0

3.2.2 MEASUREMENT INSTRUMENTS LIST

Item	Kind of Equipment	Manufacturer	Type No.	Serial No.	Calibrated until
1	Log-Bicon Antenna	SCHWARZBECK	VULB9168	VULB9168-192	10/20/2024
2	Pre-Amplifier	EM Electronics Corporation	EM330	60603	10/20/2024
3	EMI Test Receiver	R&S	ESCI	101368	10/20/2024
4	Turn Table	UC	UC3000	N/A	N/A
5	Antenna Mast	UC	UC3000	N/A	N/A

Remark: " N/A" denotes No Model No. / Serial No. and No Calibration specified.

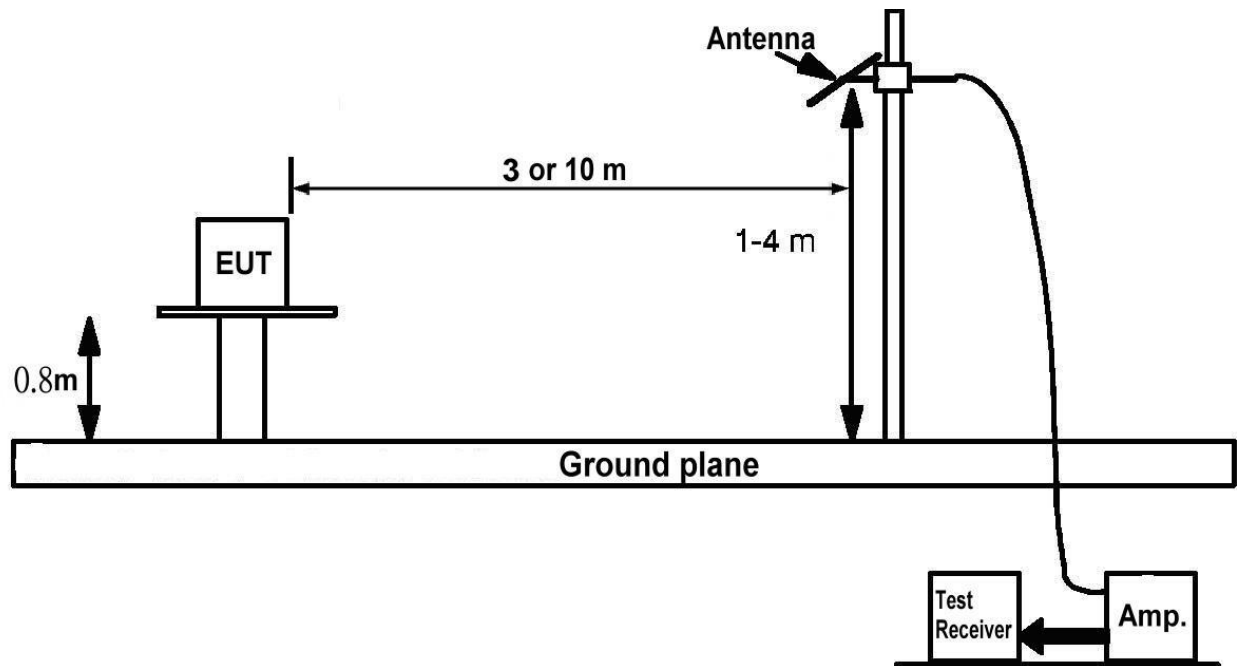
3.2.3 TEST PROCEDURE

- The measuring distance of at 3 m shall be used for measurements at frequency up to 1GHz. For frequencies above 1GHz, any suitable measuring distance may be used.
- The EUT was placed on the top of a rotating table 0.8 meters above the ground at a 3m or 10 meter open area test site. The table was rotated 360 degrees to determine the position of the highest radiation.
- The height of the equipment or of the substitution antenna shall be 0.8 m; the height of the test antenna shall vary between 1 m to 4 m. Both horizontal and vertical polarizations of the antenna are set to make the measurement.
- The initial step in collecting conducted emission data is a spectrum analyzer peak detector mode pre-scanning the measurement frequency range. Significant peaks are then marked and then Quasi Peak detector mode re-measured.
- If the Peak Mode measured value compliance with and lower than Quasi Peak Mode Limit, the EUT shall be deemed to meet QP Limits and then no additional QP Mode measurement performed.
- For the actual test configuration, please refer to the related Item –EUT Test Photos.

3.2.4 DEVIATION FROM TEST STANDARD

No deviation

3.2.5 TEST SETUP



3.2.6 EUT OPERATING CONDITIONS

The EUT tested system was configured as the statements of 2.2 Unless otherwise a special operating condition is specified in the follows during the testing.

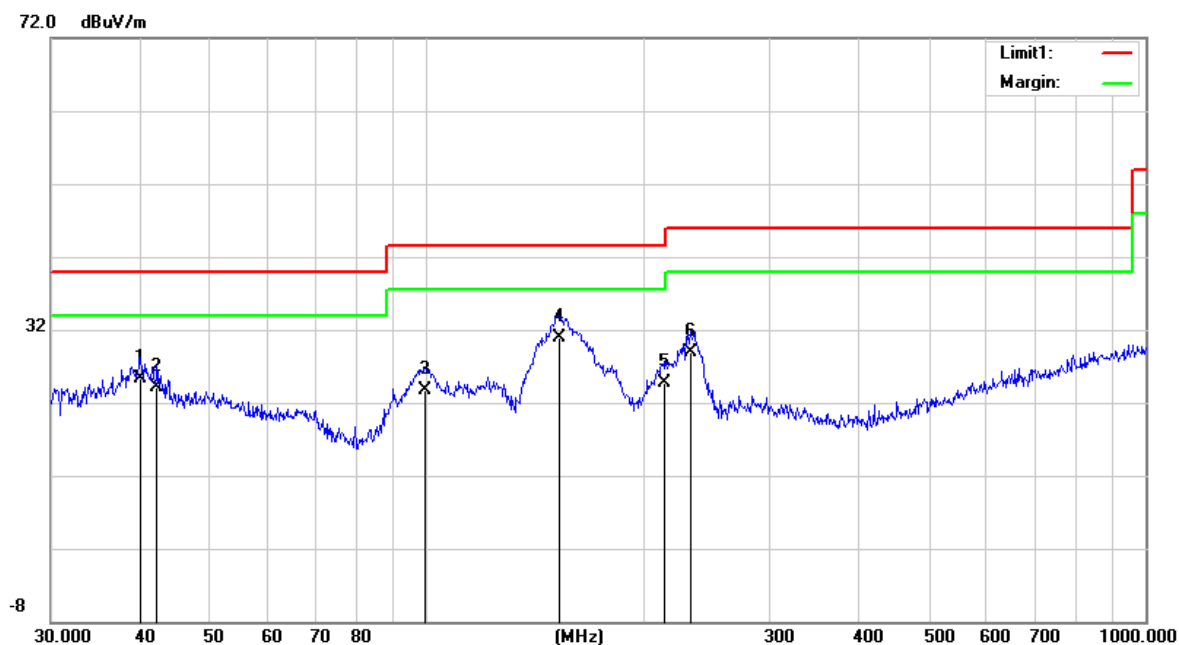
3.2.7 TEST RESULTS

EUT :	UPS uninterruptible power supply
Model No. :	UPS 1000VA Pro/800W
Test Mode :	Charging, Charging+Working, Working
Test Result:	PASS

Remark :

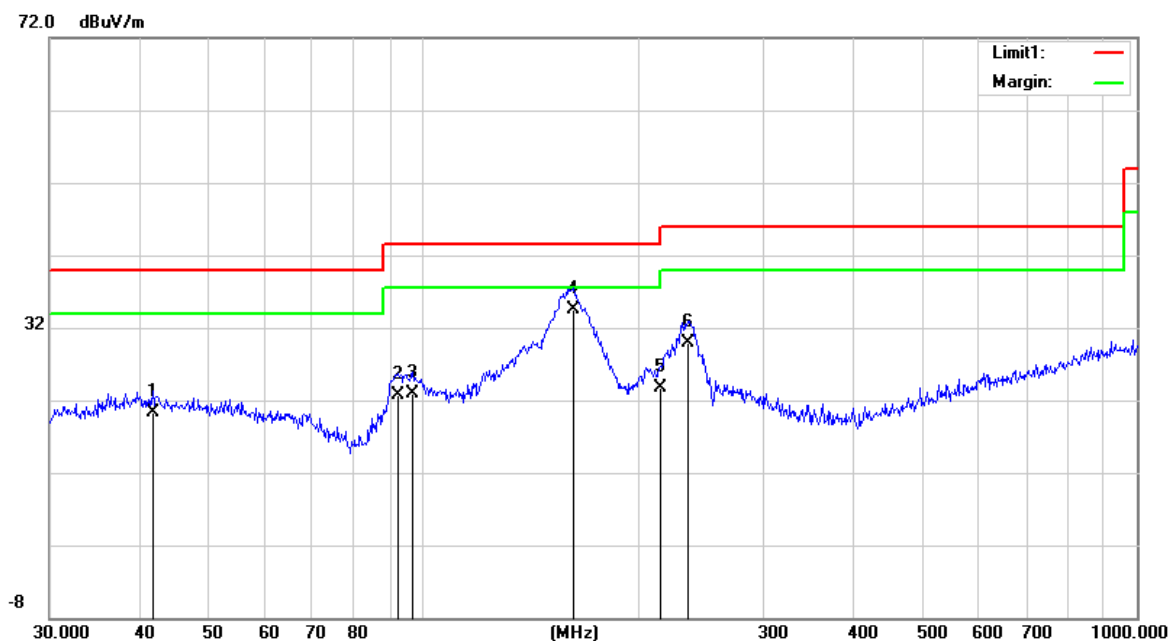
- (1) Reading in which marked as QP or Peak means measurements by using are Quasi-Peak Mode or Peak Mode with Detector BW=120KHz ; SPA setting in RBW=120KHz, VBW =120KHz, Sweep. Time = 0.3 sec./MHz.
- (2) All readings are Peak unless otherwise stated QP in column of 『Note』 . Peak denotes that the Peak reading compliance with the QP Limits and then QP Mode measurement didn't perform.
- (3) Measuring frequency range from 30MHz to 1000MHz.
- (4) If the peak scan value lower limit more than 20dB, then this signal data does not show in table.
- (5) Measurement result=Reading + Correct.

EUT :	UPS uninterruptible power supply	Model No. :	UPS 1000VA Pro/800W
Temperature :	24 °C	Relative Humidity :	55 %
Distance:	3m	Test Power :	120Vac, 60Hz
Polarization:	Vertical	Test By:	Devil
Standard:	(RE)FCC PART 15 class B 3m		
Test Mode :	Charging		



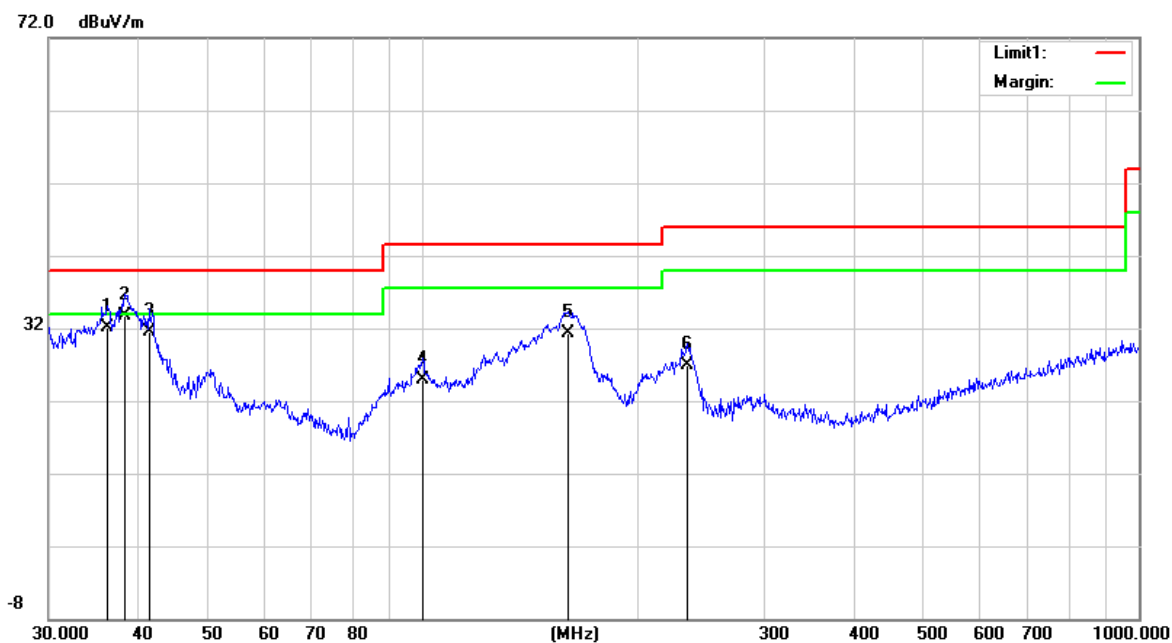
No.	Mk.	Freq.	Reading	Correct	Measure-	Limit	Over		
		MHz	Level	Factor	ment			Detector	Comment
			dBuV	dB	dBuV/m	dB/m	dB		
1		39.9941	7.72	17.50	25.22	40.00	-14.78	QP	
2		42.1542	6.80	17.39	24.19	40.00	-15.81	QP	
3		99.5279	10.64	13.06	23.70	43.50	-19.80	QP	
4	*	152.6640	14.44	16.55	30.99	43.50	-12.51	QP	
5		213.7633	10.81	13.88	24.69	43.50	-18.81	QP	
6		232.5318	14.08	14.85	28.93	46.00	-17.07	QP	

EUT :	UPS uninterruptible power supply	Model No. :	UPS 1000VA Pro/800W
Temperature :	24 °C	Relative Humidity :	55 %
Distance:	3m	Test Power :	120Vac, 60Hz
Polarization:	Horizontal	Test By:	Devil
Standard:	(RE)FCC PART 15 class B 3m		
Test Mode :	Charging		



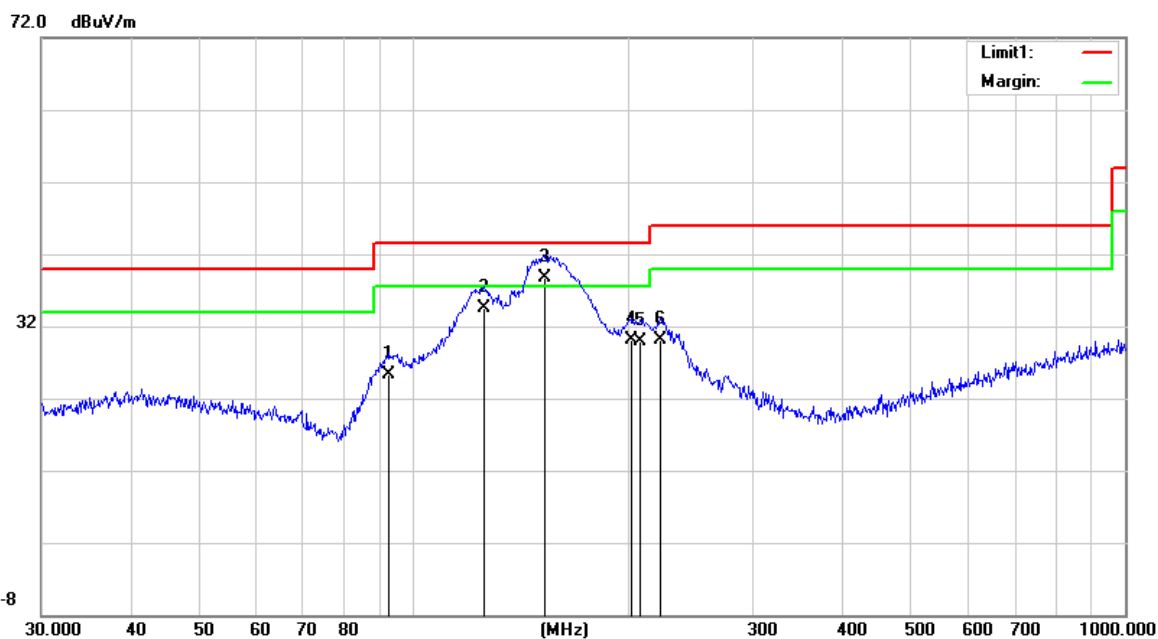
No.	Mk.	Freq.	Reading Level	Correct Factor	Measurement	Limit	Over		
		MHz	dBuV	dB	dBuV/m	dB/m	dB	Detector	Comment
1		41.8596	2.80	17.41	20.21	40.00	-19.79	QP	
2		92.1388	10.31	12.47	22.78	43.50	-20.72	QP	
3		96.7749	10.02	12.84	22.86	43.50	-20.64	QP	
4	*	162.6106	18.03	16.44	34.47	43.50	-9.03	QP	
5		215.2677	9.87	13.93	23.80	43.50	-19.70	QP	
6		234.9909	14.86	15.00	29.86	46.00	-16.14	QP	

EUT :	UPS uninterruptible power supply	Model No. :	UPS 1000VA Pro/800W
Temperature :	24 °C	Relative Humidity :	55 %
Distance:	3m	Test Power :	120Vac, 60Hz
Polarization:	Vertical	Test By:	Devil
Standard:	(RE)FCC PART 15 class B 3m		
Test Mode :	Charging+Working		



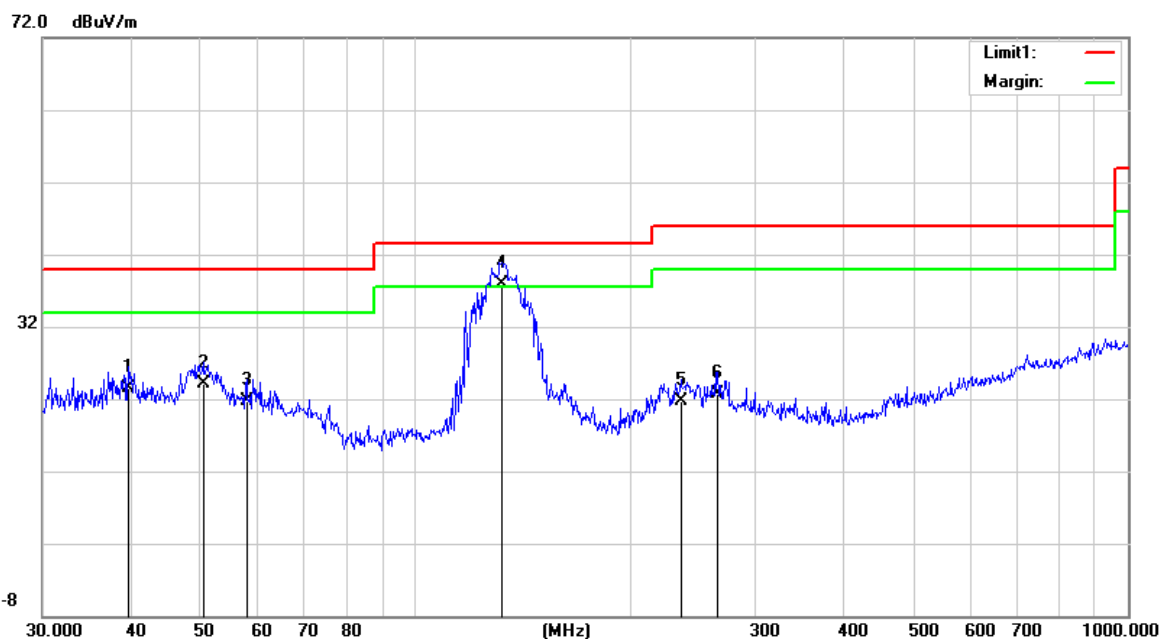
No.	Mk.	Freq.	Reading Level	Correct Factor	Measure-ment	Limit	Over		
		MHz	dBuV	dB	dBuV/m	dB/m	dB	Detector	Comment
1		36.2541	15.09	16.98	32.07	40.00	-7.93	QP	
2	*	38.3462	16.38	17.27	33.65	40.00	-6.35	QP	
3		41.4215	14.17	17.43	31.60	40.00	-8.40	QP	
4		99.8777	11.78	13.09	24.87	43.50	-18.63	QP	
5		159.7844	14.68	16.70	31.38	43.50	-12.12	QP	
6		234.1683	11.88	14.95	26.83	46.00	-19.17	QP	

EUT :	UPS uninterruptible power supply	Model No. :	UPS 1000VA Pro/800W
Temperature :	24 °C	Relative Humidity :	55 %
Distance:	3m	Test Power :	120Vac, 60Hz
Polarization:	Horizontal	Test By:	Devil
Standard:	(RE)FCC PART 15 class B 3m		
Test Mode :	Charging+Working		



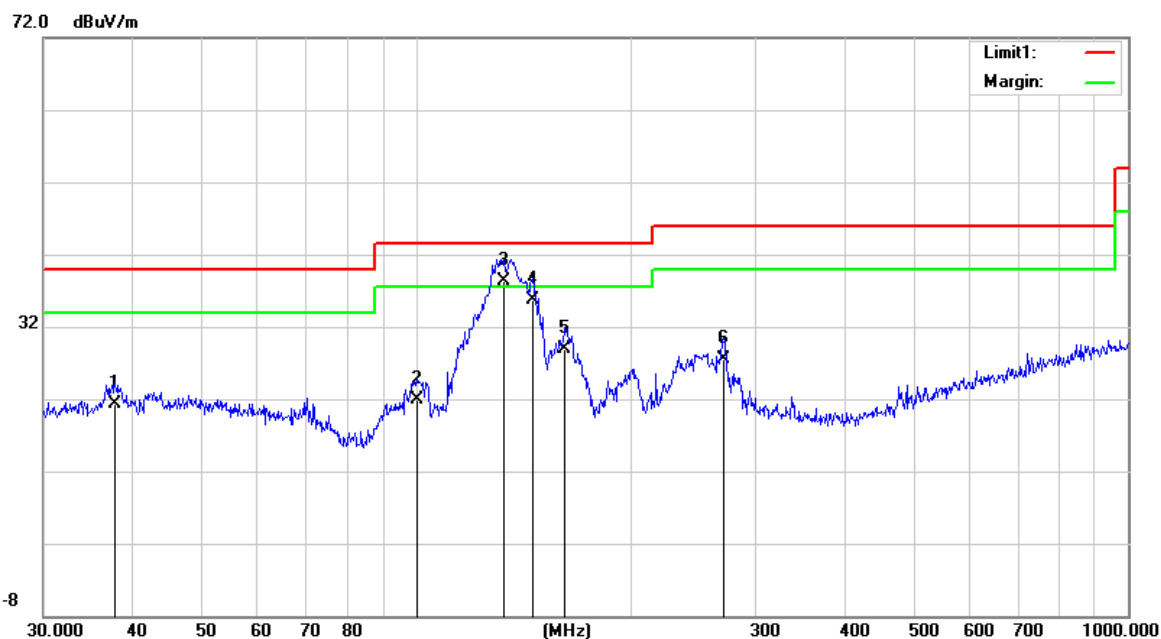
No. Mk.	Freq. MHz	Reading Level dBuV	Correct Factor dB	Measure- ment dBuV/m	Limit dB/m	Over dB	Detector	Comment
1	92.4624	12.72	12.50	25.22	43.50	-18.28	QP	
2	125.4457	19.17	15.35	34.52	43.50	-8.98	QP	
3 *	152.6641	22.15	16.55	38.70	43.50	-4.80	QP	
4	202.8103	16.59	13.50	30.09	43.50	-13.41	QP	
5	208.5801	16.26	13.70	29.96	43.50	-13.54	QP	
6	222.1698	15.78	14.23	30.01	46.00	-15.99	QP	

EUT :	UPS uninterruptible power supply	Model No. :	UPS 1000VA Pro/800W
Temperature :	24 °C	Relative Humidity :	55 %
Distance:	3m	Test Power :	120Vac, 60Hz
Polarization:	Vertical	Test By:	Devil
Standard:	(RE)FCC PART 15 class B 3m		
Test Mode :	Working		



No.	Mk.	Freq.	Reading Level	Correct Factor	Measurement	Limit	Over		
		MHz	dBuV	dB	dBuV/m	dB/m	dB	Detector	Comment
1		39.5756	6.06	17.44	23.50	40.00	-16.50	QP	
2		50.4089	7.08	16.96	24.04	40.00	-15.96	QP	
3		57.9992	5.66	16.12	21.78	40.00	-18.22	QP	
4	*	132.2205	22.19	15.79	37.98	43.50	-5.52	QP	
5		235.8163	6.72	15.05	21.77	46.00	-24.23	QP	
6		265.6757	6.77	15.96	22.73	46.00	-23.27	QP	

EUT :	UPS uninterruptible power supply	Model No. :	UPS 1000VA Pro/800W
Temperature :	24 °C	Relative Humidity :	55 %
Distance:	3m	Test Power :	120Vac, 60Hz
Polarization:	Horizontal	Test By:	Devil
Standard:	(RE)FCC PART 15 class B 3m		
Test Mode :	Working		



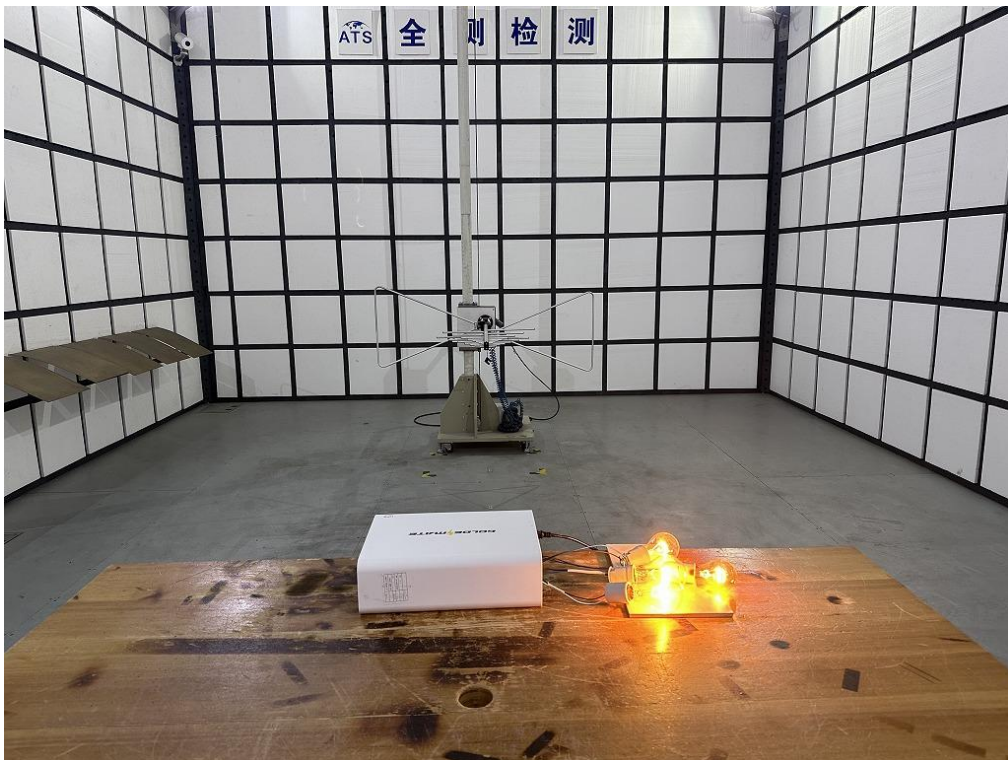
No. Mk.	Freq.	Reading Level	Correct Factor	Measure-ment	Limit	Over		
	MHz	dBuV	dB	dBuV/m	dB/m	dB	Detector	Comment
1	37.8121	4.04	17.19	21.23	40.00	-18.77	QP	
2	100.5806	8.79	13.15	21.94	43.50	-21.56	QP	
3 *	133.1511	22.36	15.85	38.21	43.50	-5.29	QP	
4	145.8609	19.27	16.42	35.69	43.50	-7.81	QP	
5	162.0414	12.49	16.50	28.99	43.50	-14.51	QP	
6	270.3747	11.27	16.17	27.44	46.00	-18.56	QP	

4. EUT TEST PHOTOS

Conducted Measurement Photo



Radiated Measurement Photo



5. EUT PHOTOS



END OF REPORT